VISUALIZATION & DASHBOARDS

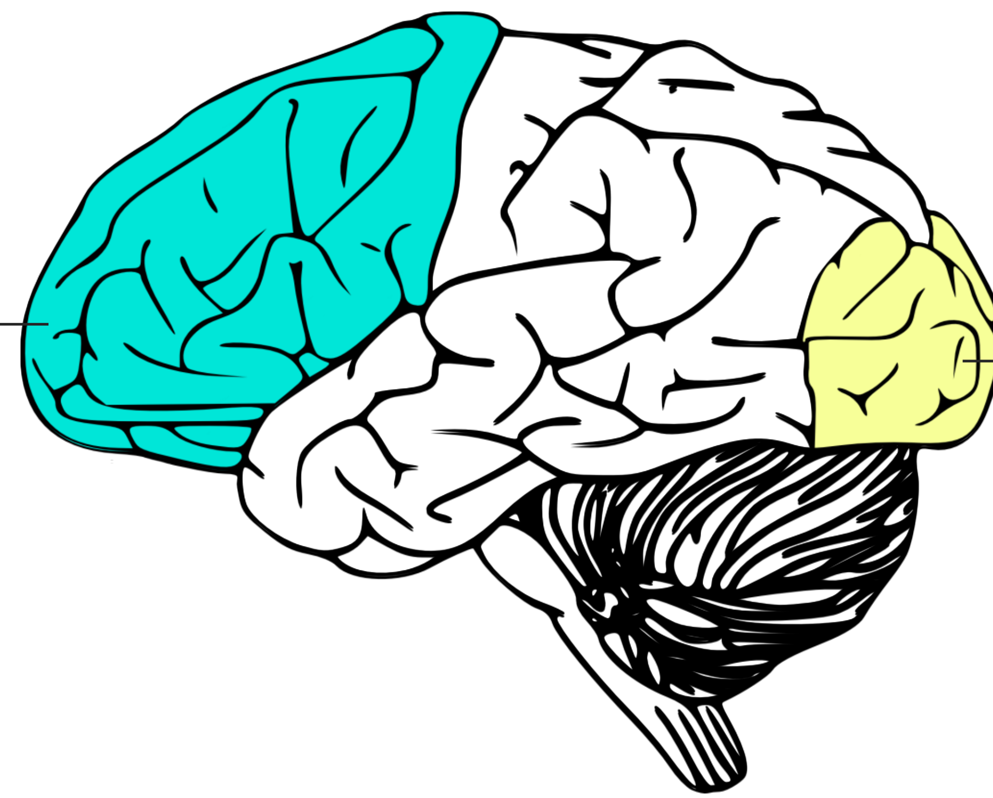
*Copyright Maven Analytics, LLC

BRING YOUR DATA TO LIFE

The human brain isn't built to interpret raw data; we need **clear patterns** and **visual cues** to help us quickly make sense of complex information

Prefrontal Cortex

- *Located in the frontal lobe*
- *Responsible for cognitive functioning & problem solving*
- *Slow & conscious*
- *Helps us make sense of non-visual information (like raw data)*



Visual Cortex

- *Located in the occipital lobe*
- *Responsible for visual perception & understanding*
- *Instantaneous & subconscious*
- *Helps us make sense of colors, patterns, shapes, sizes, etc.*

Data visualization puts both our prefrontal and visual cortex to work, combining the power of **cognition** *(slow and conscious)* and **perception** *(instantaneous)*

*Copyright Maven Analytics, LLC

DATA VISUALIZATION

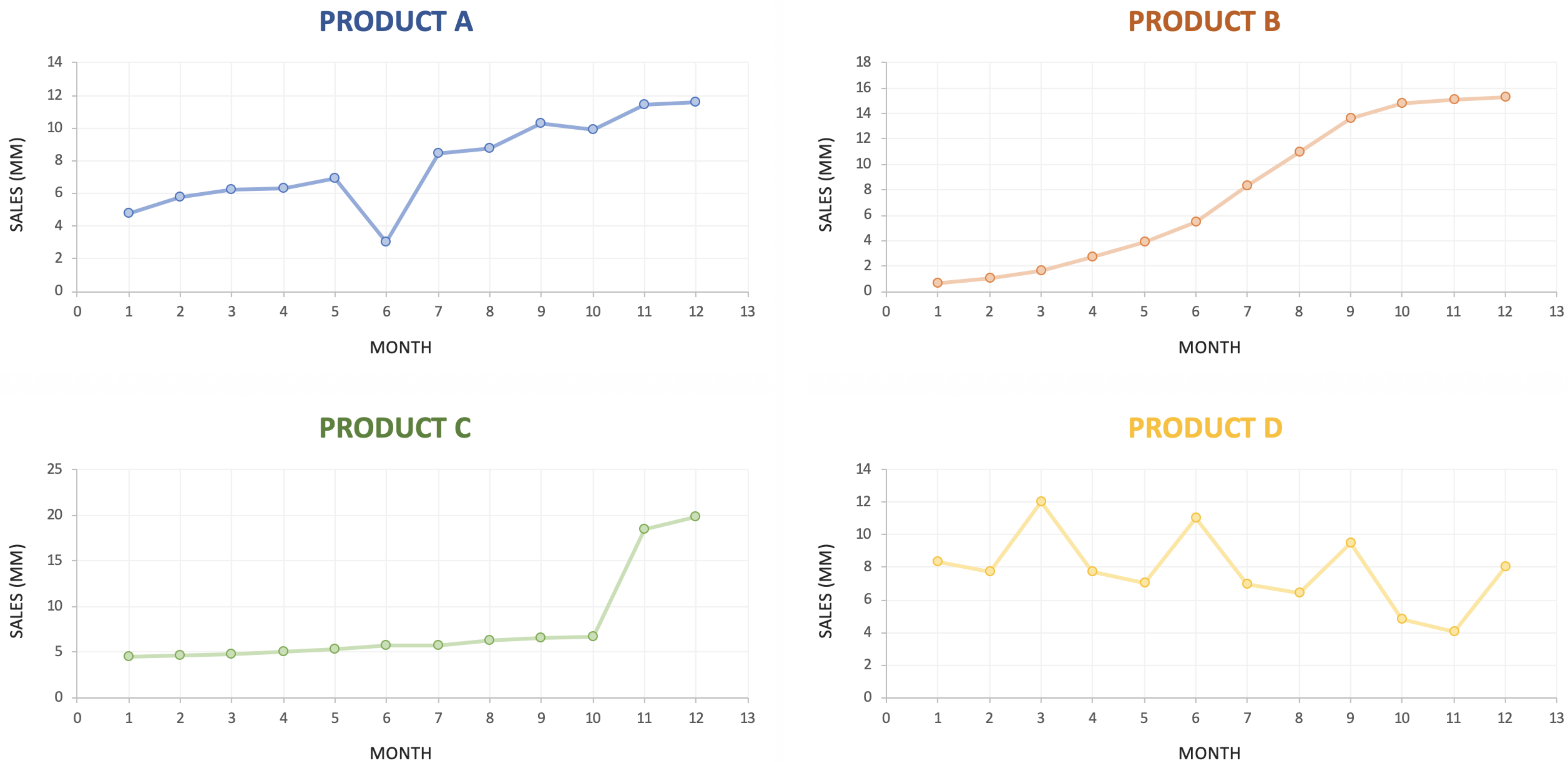
In **10 seconds**, what can you learn from the data below?

Product A		Product B		Product C		Product D	
Month	Sales (MM)	Month	Sales (MM)	Month	Sales (MM)	Month	Sales (MM)
1	4.80	1	0.67	1	4.53	1	8.35
2	5.78	2	1.05	2	4.61	2	7.72
3	6.24	3	1.62	3	4.74	3	12.05
4	6.34	4	2.67	4	5.10	4	7.70
5	6.95	5	3.91	5	5.32	5	7.05
6	3.02	6	5.49	6	5.70	6	11.05
7	8.45	7	8.36	7	5.77	7	6.95
8	8.79	8	10.99	8	6.32	8	6.39
9	10.30	9	13.58	9	6.56	9	9.50
10	9.93	10	14.81	10	6.64	10	4.83
11	11.40	11	15.13	11	18.50	11	4.03
12	11.56	12	15.26	12	19.80	12	8.03

*Copyright Maven Analytics, LLC

DATA VISUALIZATION

How about now?



This is a variation of ***Anscombe's Quartet***

Despite sharing similar descriptive stats, each series tells a very different visual story

*Copyright Maven Analytics, LLC

3 KEY QUESTIONS

1 What **TYPE OF DATA** are you working with?

- Geospatial? Time-series? Hierarchical? Financial?

2 What do you want to **COMMUNICATE**?

- Comparison? Composition? Relationship? Distribution?

3 Who is the **END USER** and what do they need?

- Analyst? Manager? Executive? General public?

*Copyright Maven Analytics, LLC

3 KEY QUESTIONS

1 What **TYPE OF DATA** are you working with?

- Time-series
- Financial
- Geospatial
- Textual
- Categorical
- Funnel
- Hierarchical
- Survey

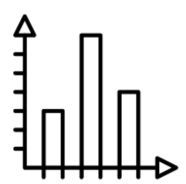
The type of data you're working with often determines **which type of visual will best represent it**; for example, using maps to represent geospatial data, line charts for time-series data, or tree maps for hierarchical data

*Copyright Maven Analytics, LLC

3 KEY QUESTIONS

2 What do you want to **COMMUNICATE**?

COMPARISON

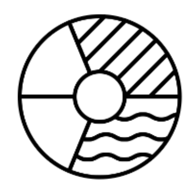


Used to **compare values over time or across categories**

Common visuals:

- Column Chart
- Bar Chart
- Clustered Column/Bar
- Data Table/Heat Map
- Radar Chart
- Line Chart *(time series)*
- Area Chart *(time series)*

COMPOSITION

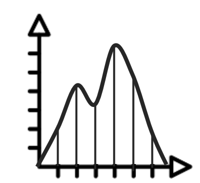


Used to **break down the component parts of a whole**

Common visuals:

- Stacked Bar/Column Chart
- Pie/Donut Chart
- Stacked Area *(time series)*
- Waterfall Chart *(gains/losses)*
- Funnel Chart *(sequential stages)*
- Tree Map/sunburst *(hierarchies)*
- Map/Choropleth *(geospatial)*

DISTRIBUTION

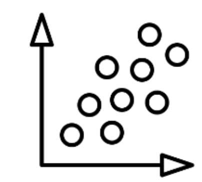


Used to **show the frequency of values within a series**

Common visuals:

- Histogram
- Density Plot
- Box & Whisker
- Violin Plot
- Scatter Plot
- Data Table/Heat Map
- Map/Choropleth *(geospatial)*

RELATIONSHIP



Used to **show correlation between multiple variables**

Common visuals:

- Scatter Plot
- Bubble Chart
- Data Table/Heat Map
- Correlation Matrix

There are ***hundreds* of charts to choose from**, but most either serve specialized purposes *(like stock charts, Gantt charts and network diagrams)* or are variations of these common types *(gauges, bullet charts, chord diagrams, etc.)*

**Copyright Maven Analytics, LLC*

3 KEY QUESTIONS

3 Who is the **END USER** and what do they need?

THE **ANALYST**

Likes to see details and understand exactly what's happening at a granular level

- Tables or combo charts
- Granular detail to support root-cause analysis

THE **MANAGER**

Wants summarized data with clear, actionable insights to help operate the business

- Common charts & graphs
- Some detail, but only when it supports a specific insight

THE **EXECUTIVE**

Needs high-level, crystal clear KPIs to track business health and topline performance

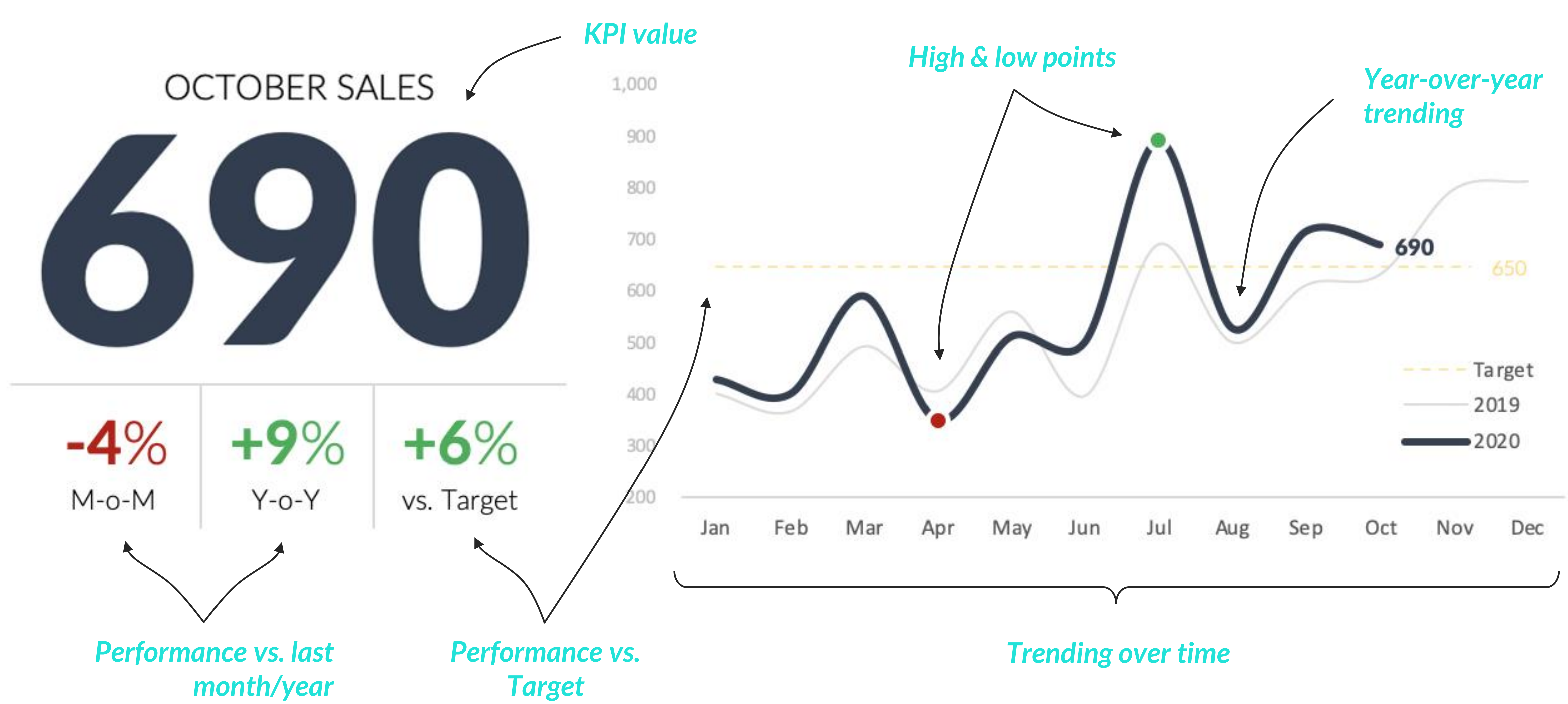
- KPI cards or simple charts
- Minimal detail, unless it adds critical context to KPIs

How you visualize and present your data is a function of **who will be consuming it**; a fellow analyst may want to see granular details, while managers and executives often prefer topline KPIs and clear, data-driven insights

*Copyright Maven Analytics, LLC

CONTEXT IS KEY

When you are presenting or visualizing data, remember that **context is key**; context gives numbers *meaning*, and helps users interpret them accurately



- We drove **690 sales** in October
- Sales are **down 4%** vs. last month, but **up 9%** year-over-year
- We **exceeded our target** in Sep/Oct after falling short in August
- We see an **upward trend** in 2020, with a significant **July peak**
- Based on seasonality, we can expect a **strong Nov/Dec**

*Copyright Maven Analytics, LLC

ANALYTICS DASHBOARDS

Dashboards are analytics tools designed to consolidate data from multiple sources, track key metrics at a glance, and facilitate data-driven decision making

Centralized, user-facing dashboard

*Copyright Maven Analytics, LLC

CASE STUDY: DATA VISUALIZATION

THE
SITUATION

You've just been hired as Lead Business Intelligence Analyst for **Maven Toys**, a nationwide chain of toy stores.

THE
ASSIGNMENT

Your assignment is to **design a dashboard for Regional Sales Managers**, to help them track revenue trends by region.

They review the dashboard once a month, and need information on sales trends, product performance, and lost revenue due to inventory shortages.

THE
OBJECTIVES

1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

MAVEN TOYS

*Copyright Maven Analytics, LLC

DASHBOARD DESIGN PROCESS

1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

Dashboards can be built to serve **many purposes**, including executive-level reporting, performance deep dives, exploratory analysis, or infographic-style storytelling.

How you design your dashboard is largely a function of the *purpose* it will serve, and the *audience* it will be serving.

Key questions to consider:

- *Who will be the **end-users** of your dashboard?*
- *What are their **key business goals** and objectives?*
- *What are the **most important questions** they need answers to?*
- *How **frequently** will the dashboard be reviewed?*

*Copyright Maven Analytics, LLC

STEP 1: DEFINE THE PURPOSE

1. Who will be the end-users of your dashboard?

2. What are their key business goals and objectives?

3. What are the most important questions they need answers to?

4. How frequently will the dashboard be reviewed?

*Copyright Maven Analytics, LLC

STEP 1: DEFINE THE PURPOSE

1. Who will be the end-users of your dashboard?
   - ***Regional sales managers***

2. What are their key business goals and objectives?
   - ***Increase revenue, minimize revenue loss due to lack of inventory***

3. What are the most important questions they need answers to?
   - ***Revenue going up/down? Main revenue drivers? Missing any products in stock?***

4. How frequently will the dashboard be reviewed?
   - ***Once a month***

*Copyright Maven Analytics, LLC

DASHBOARD DESIGN PROCESS

1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

Once you've defined the purpose of your dashboard, it's important to identify **which metrics and KPIs** to include.

Focus on the metrics which *directly* align with key business goals, and consider the level of detail most appropriate for your audience.

Key questions to consider:

- *Which metrics **accurately measure** each business goal?*
- *What **level of detail** is appropriate for each metric?*

PRO TIP: You might be tempted to include *everything* in your dashboard, but remember less is more; focus on the metrics that *matter*!

*Copyright Maven Analytics, LLC

STEP 2: CHOOSE THE RIGHT METRICS

Business Goal:	Metrics:	Level of Detail:
Increase revenue		
Minimize revenue loss due to lack of inventory		

*Copyright Maven Analytics, LLC

STEP 2: CHOOSE THE RIGHT METRICS

Business Goal:	Metrics:	Level of Detail:
Increase revenue	**Total Revenue**	Date, Region, Category, Product
	MoM Revenue Change	Product
	MoM % Revenue Change	Region, Category
	YoY % Revenue Change	Region
Minimize revenue loss due to lack of inventory	**Stock on Hand**	Store-Product
	Est Monthly Revenue Loss*	Region

** Based on average monthly revenue for out-of-stock products*

*Copyright Maven Analytics, LLC

DASHBOARD DESIGN PROCESS

1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

Dashboards are all about communicating information **quickly and clearly**.

Remember to use charts and visuals suited to the type of data you're working with, the story you're communicating, and the end user consuming the information.

Want to allow for some exploratory analysis? **Add filters and interactivity** to allow users to explore on their own, answer new questions, and discover fresh insights.

PRO TIP: One of the most common mistakes we see is prioritizing *variety* over effectiveness. Always choose the right chart for the job!

*Copyright Maven Analytics, LLC

STEP 3: PRESENT THE DATA EFFECTIVELY

Metric & Level of Detail:		Visual Type:
• Total Revenue by Date	→	
• Total Revenue by Region	→	
• Total Revenue by Category	→	
• Total Revenue by Product	→	
• MoM Revenue Change by Product	→	
• MoM % Revenue Change by Region	→	
• MoM % Revenue Change by Category	→	
• YoY % Revenue Change by Region	→	
• Stock on Hand by Store-Product	→	
• Est. Monthly Revenue Loss by Region	→	

Filters:

*Copyright Maven Analytics, LLC

STEP 3: PRESENT THE DATA EFFECTIVELY

Metric & Level of Detail:		Visual Type:
Total Revenue by Date	→	**1) Line Chart**
Total Revenue by Region	→	**2) KPI Card** *(#1)*
Total Revenue by Category	→	**3) Bar Chart** *(#1)*
Total Revenue by Product	→	**4) Table** *(#1)*
MoM Revenue Change by Product	→	**5) Table** *(#1)*
MoM % Revenue Change by Region	→	**6) KPI Card** *(#2)*
MoM % Revenue Change by Category	→	**7) Bar Chart** *(#2)*
YoY % Revenue Change by Region	→	**8) KPI Card** *(#3)*
Stock on Hand by Store-Product	→	**9) Table** *(#2)*
Est. Monthly Revenue Loss by Region	→	**10) KPI Card** *(#4)*

Filters:

- **Month** *(interactive)*
- **Region** *(interactive)*
- **Top/Bottom 5 Products**
- **Store-Products with 0 stock**

This is typically an **evolving, iterative process**, so don't expect to come up with an exhaustive set of metrics and visuals off the top of your head!

*Copyright Maven Analytics, LLC

STEP 3: PRESENT THE DATA EFFECTIVELY

MAVEN TOYS

REGIONAL SALES DASHBOARD

September 2021

Region: **New York**

2 **$50,618**

Total Monthly Revenue

6 **1.6%**

M-o-M Revenue % Change

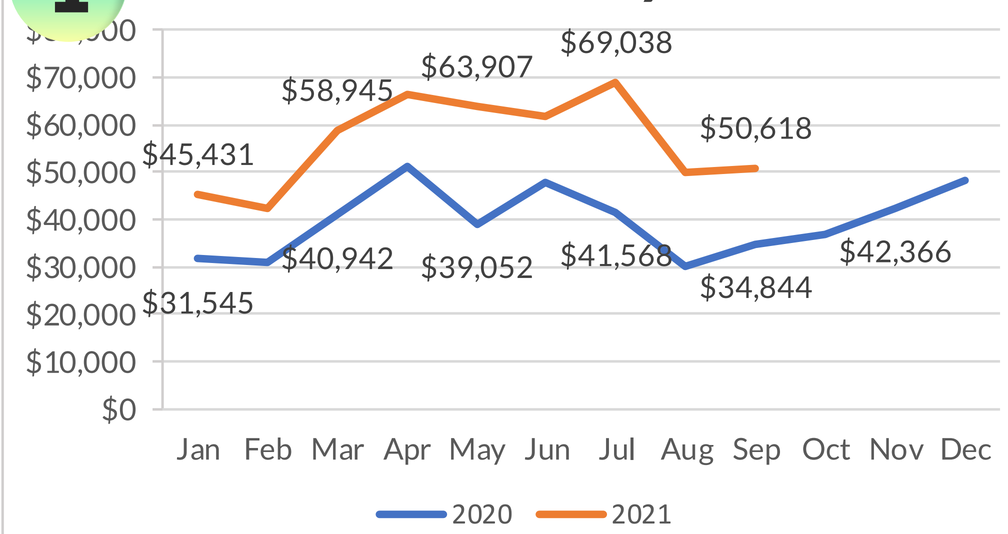
10 **$1,640**

Potential Monthly Revenue Loss

8 **45.3%**

Y-o-Y Revenue % Change

1 **Total Revenue by Date**



Products with 0 Stock by Store

9

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

Top 5 Products by MoM Revenue Change

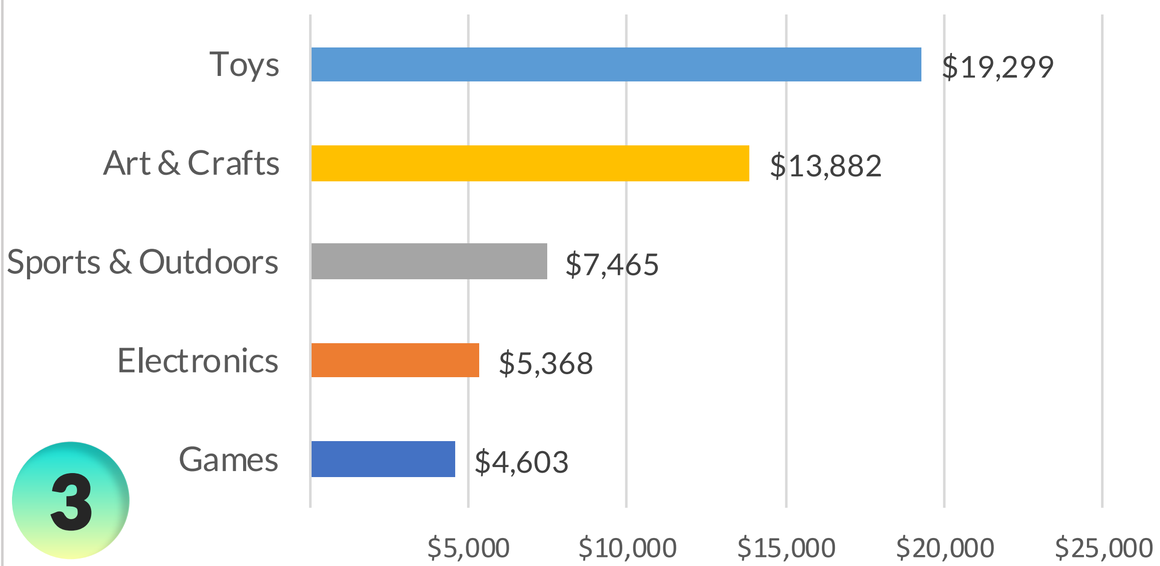
Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	$2,011
Toys	Dinosaur Figures	$2,893	$989
Games	Monopoly	$900	$740
Art & Crafts	Magic Sand	$4,589	$688
Art & Crafts	Barrel O' Slime	$1,357	$551

4 5

Bottom 5 Products by MoM Revenue Change

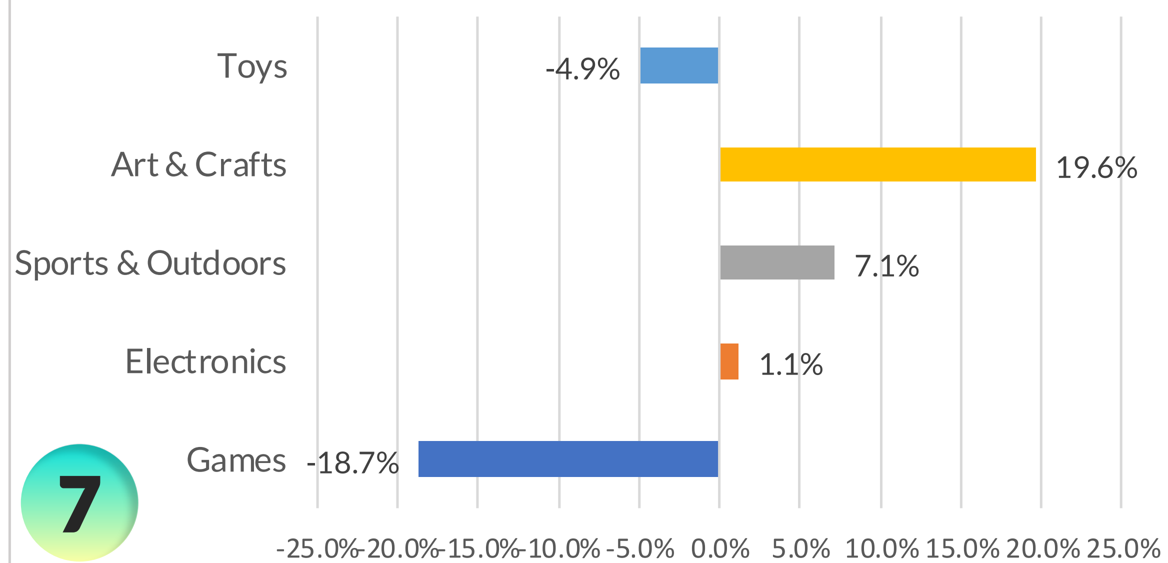
Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

Total Revenue by Product Category



3

MoM Revenue % Change by Product Category



7

*Copyright Maven Analytics, LLC

DASHBOARD DESIGN PROCESS

1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

When it comes to dashboard design, **real estate is precious**; cut anything that takes up space but doesn't add value.

Things like 3D formats, chart borders, excessive colors, or background images only distract users from the story and insights you are trying to communicate.

Clarity *always* trumps aesthetics!

> “*Perfection is achieved not when there is nothing more to add, **but when there is nothing left to take away***”
>
> *Antoine de Saint-Exupery*

*Copyright Maven Analytics, LLC

STEP 4: ELIMINATE CLUTTER & NOISE

MAVEN TOYS

REGIONAL SALES DASHBOARD

September 2021

Region: **New York**

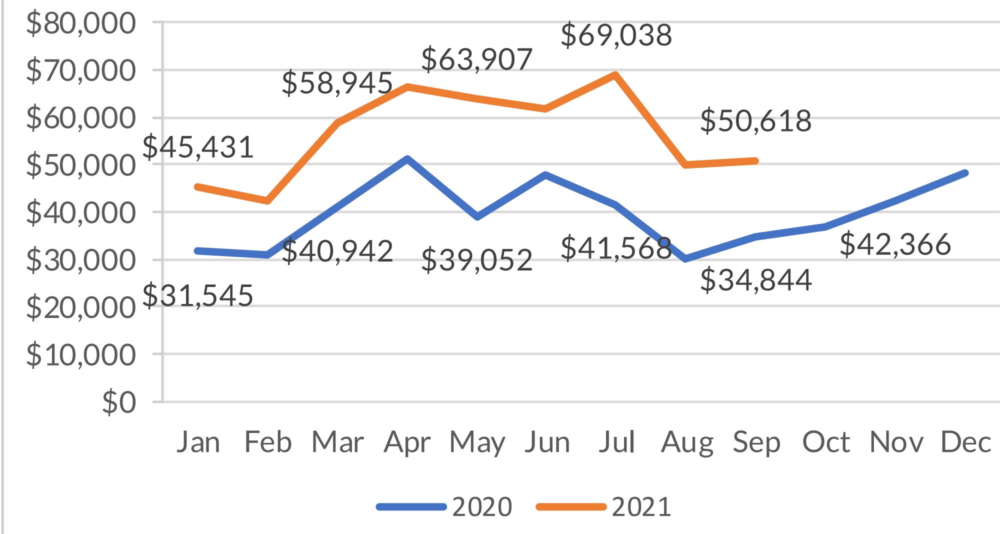
$50,618
Total Monthly Revenue

1.6%
M-o-M Revenue % Change

$1,640
Potential Monthly Revenue Loss

45.3%
Y-o-Y Revenue % Change

Total Revenue by Date



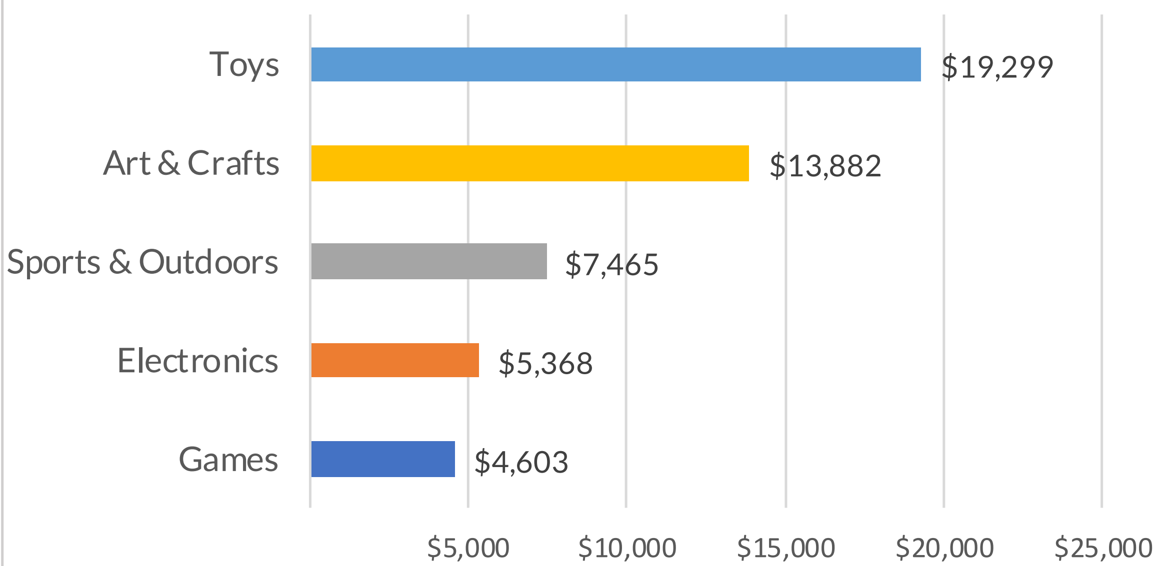
Products with 0 Stock by Store

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

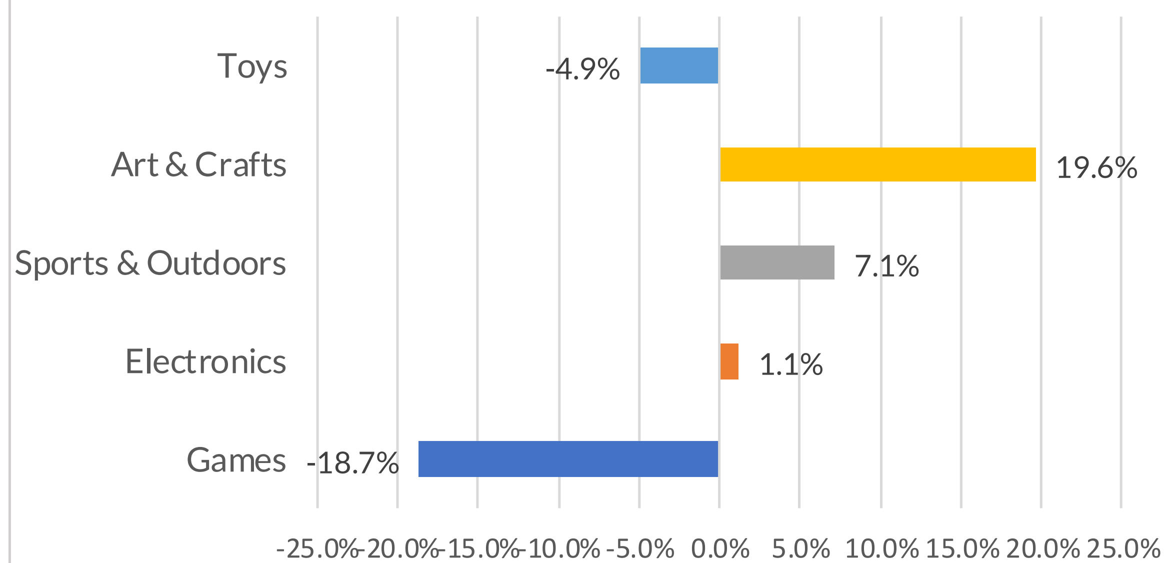
Top 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	$2,011
Toys	Dinosaur Figures	$2,893	$989
Games	Monopoly	$900	$740
Art & Crafts	Magic Sand	$4,589	$688
Art & Crafts	Barrel O' Slime	$1,357	$551

Total Revenue by Product Category



MoM Revenue % Change by Product Category



Bottom 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

*Copyright Maven Analytics, LLC

STEP 4: ELIMINATE CLUTTER & NOISE

MAVEN TOYS

REGIONAL SALES DASHBOARD

September 2021

Region: **New York**

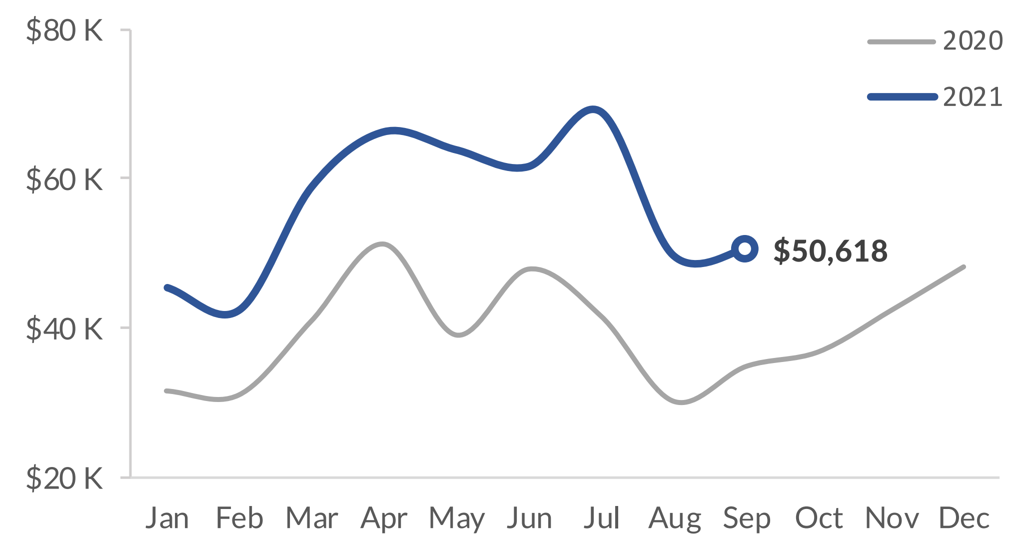
$50,618
Total Monthly Revenue

1.6%
M-o-M Revenue % Change

$1,640
Potential Monthly Revenue **Loss**

45.3%
Y-o-Y Revenue % Change

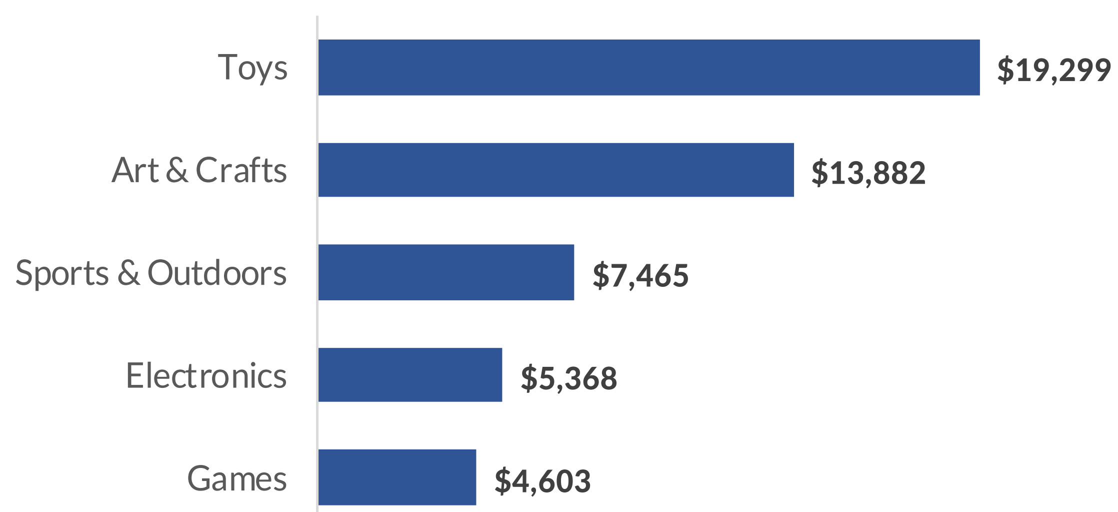
Total Revenue by Date



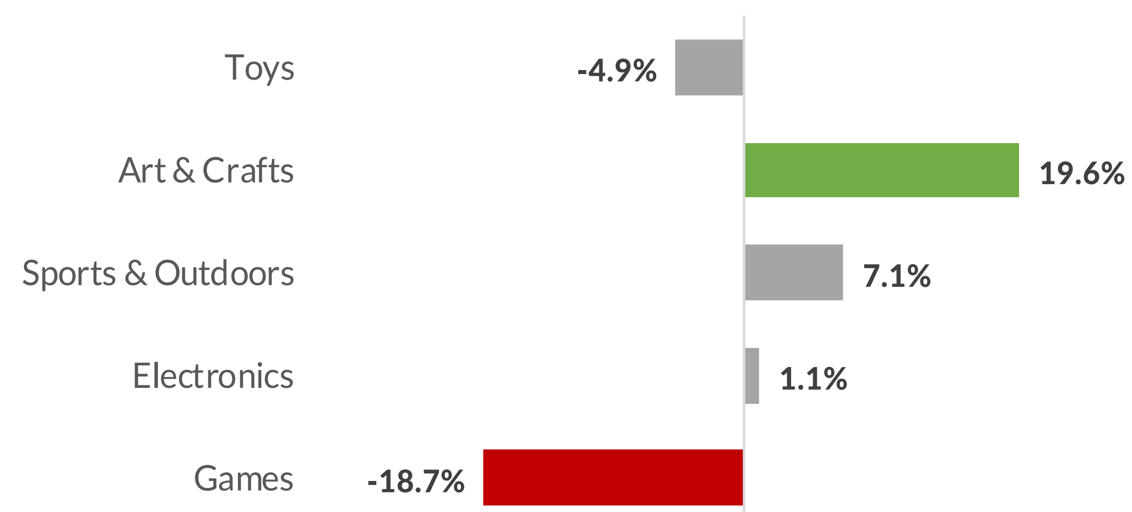
Products with 0 Stock by Store

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

Total Revenue by Product Category



MoM Revenue % Change by Product Category



Top 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	$2,011
Toys	Dinosaur Figures	$2,893	$989
Games	Monopoly	$900	$740
Art & Crafts	Magic Sand	$4,589	$688
Art & Crafts	Barrel O' Slime	$1,357	$551

Bottom 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

*Copyright Maven Analytics, LLC

DASHBOARD DESIGN PROCESS

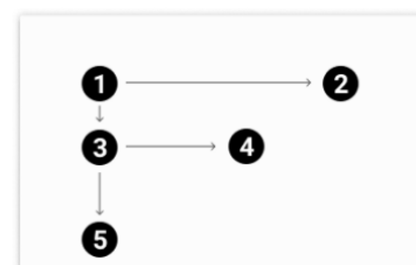
1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

A good layout features the most important insights and trends upfront, and guides the viewer through a logical story; **don't expect users to connect the dots on their own!**

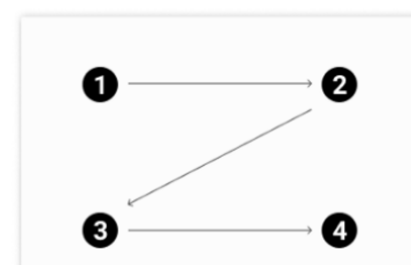
Tips to focus attention:

- *Use **pre-attentive attributes** like size, color, and position to highlight key data points or specific patterns*
- *Use **Gestalt principles** like proximity, enclosure and connection to group related visual elements*
- *Consider common **reading patterns** (like Z or F patterns) when designing your dashboard layout*

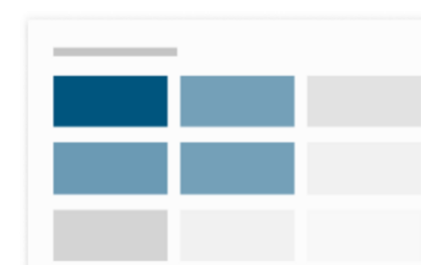
F-PATTERN



Z-PATTERN



MOST ATTRACTING AREAS



*Copyright Maven Analytics, LLC

STEP 4: ELIMINATE CLUTTER & NOISE

MAVEN TOYS

REGIONAL SALES DASHBOARD

September 2021

Region: **New York**

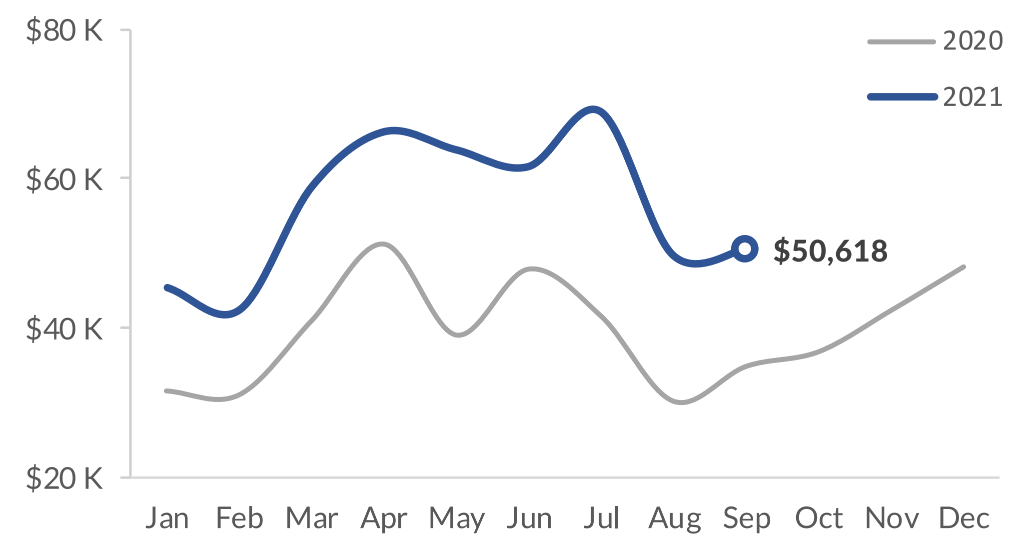
$50,618
Total Monthly Revenue

1.6%
M-o-M Revenue % Change

$1,640
Potential Monthly Revenue **Loss**

45.3%
Y-o-Y Revenue % Change

Total Revenue by Date



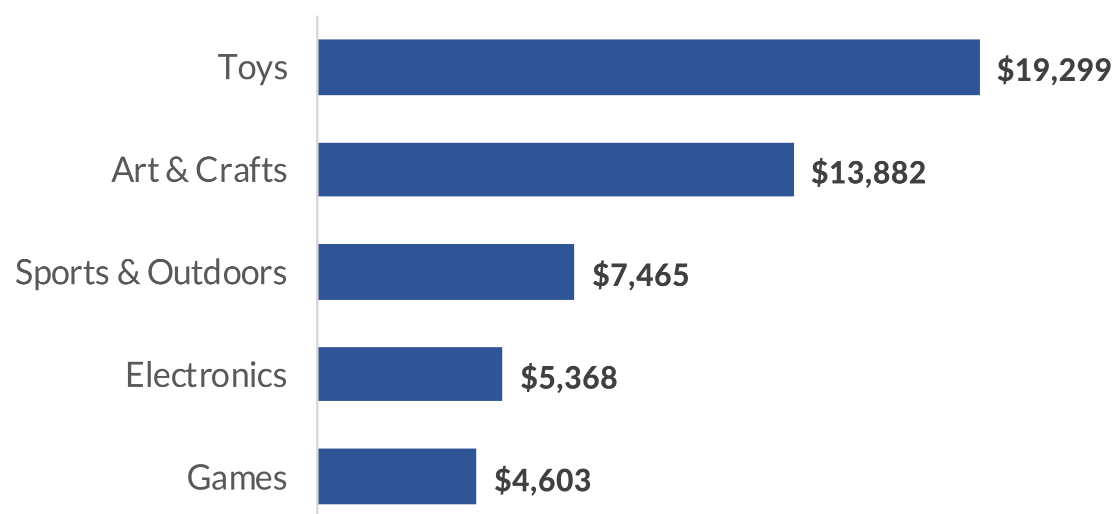
Products with 0 Stock by Store

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

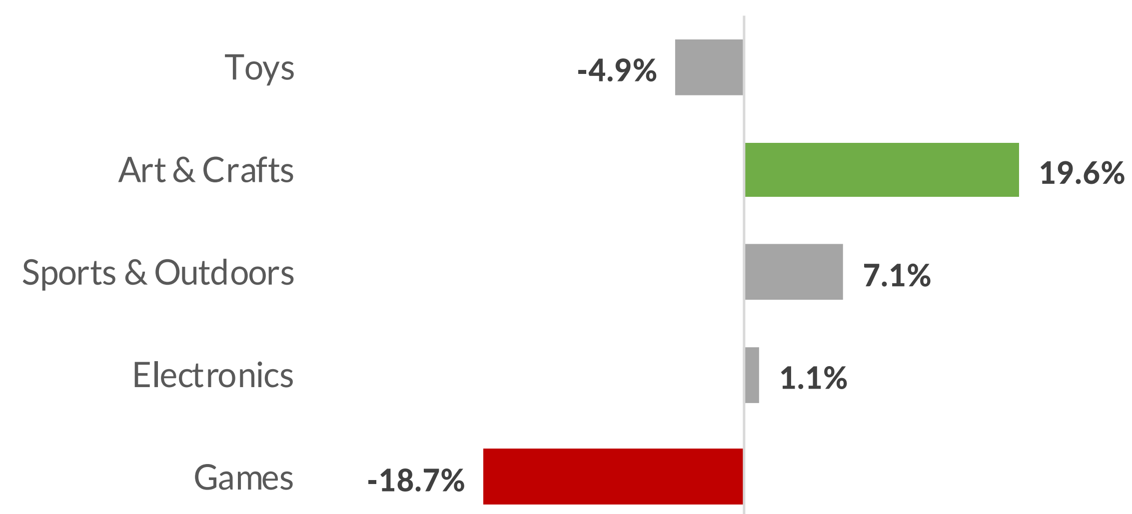
Top 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	$2,011
Toys	Dinosaur Figures	$2,893	$989
Games	Monopoly	$900	$740
Art & Crafts	Magic Sand	$4,589	$688
Art & Crafts	Barrel O' Slime	$1,357	$551

Total Revenue by Product Category



MoM Revenue % Change by Product Category



Bottom 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

*Copyright Maven Analytics, LLC

STEP 5: USE LAYOUT TO FOCUS ATTENTION

REGIONAL SALES DASHBOARD

September 2021

Region: **New York**

MAVEN TOYS

Total Revenue & Revenue % Changes

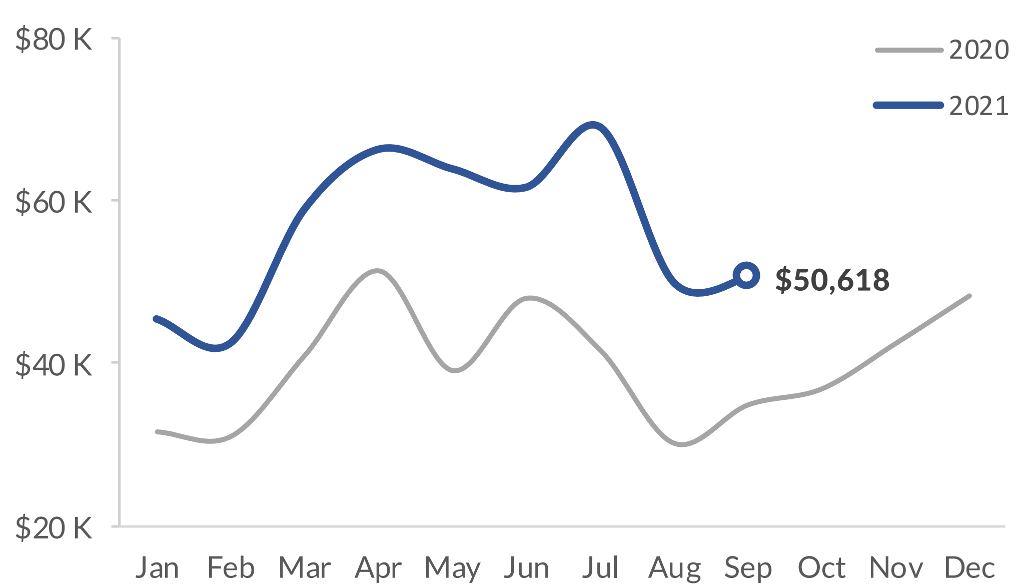
$50,618

Total Monthly Revenue

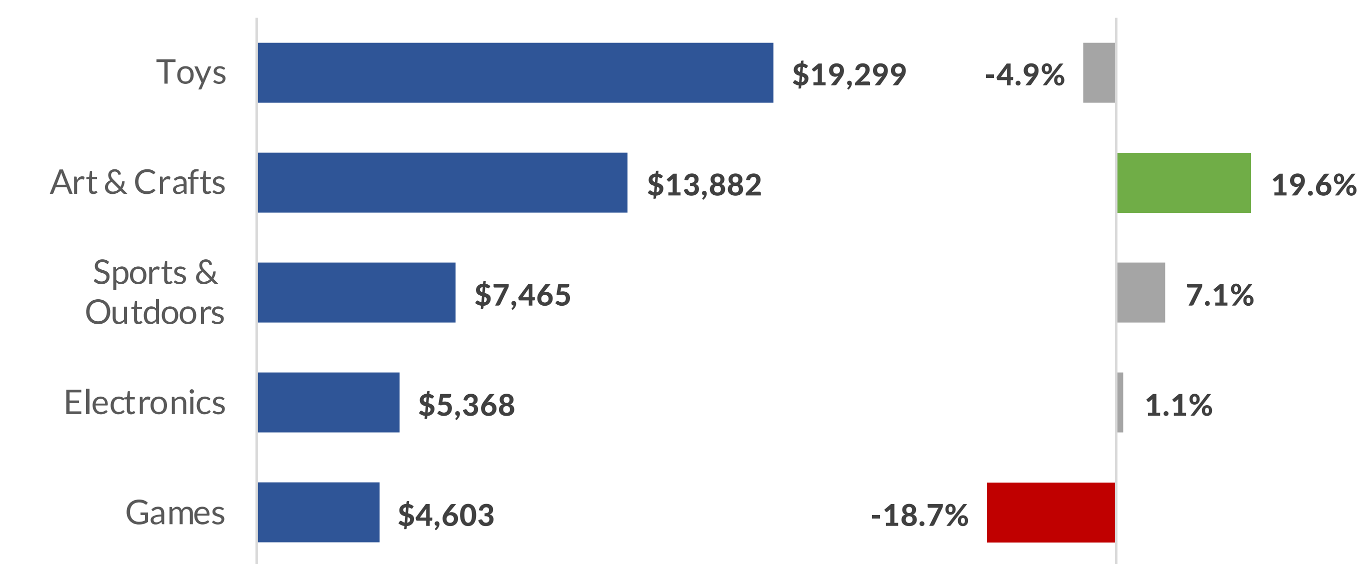
1.6%
M-o-M

45.3%
Y-o-Y

Total Revenue by Date



Revenue & MoM Revenue % Change by Product Category



Products with 0 Stock

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

$1,640

Potential Monthly Revenue **Loss**

Top & Bottom 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	$2,011
Toys	Dinosaur Figures	$2,893	$989
Games	Monopoly	$900	$740
Art & Crafts	Magic Sand	$4,589	$688
Art & Crafts	Barrel O' Slime	$1,357	$551

Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

*Copyright Maven Analytics, LLC

DASHBOARD DESIGN PROCESS

1. Define the purpose
2. Choose the right metrics
3. Present the data effectively
4. Eliminate clutter & noise
5. Use layout to focus attention
6. Tell a clear story

Human beings aren't inspired by numbers, charts or graphs; **we're inspired by stories**.

The most effective BI dashboards use data to drive real-world business outcomes; they tell clear, data-driven stories designed to expose key insights and inspire stakeholders to act.

If your dashboard doesn't inspire action or facilitate change, ***what purpose does it serve***?

PRO TIP: Don't be afraid of a little text! Using descriptive titles and data labels can be a great way to create narratives within your dashboard

*Copyright Maven Analytics, LLC

STEP 5: USE LAYOUT TO FOCUS ATTENTION

REGIONAL SALES DASHBOARD

September 2021

Region: **New York**

MAVEN TOYS

Total Revenue & Revenue % Changes

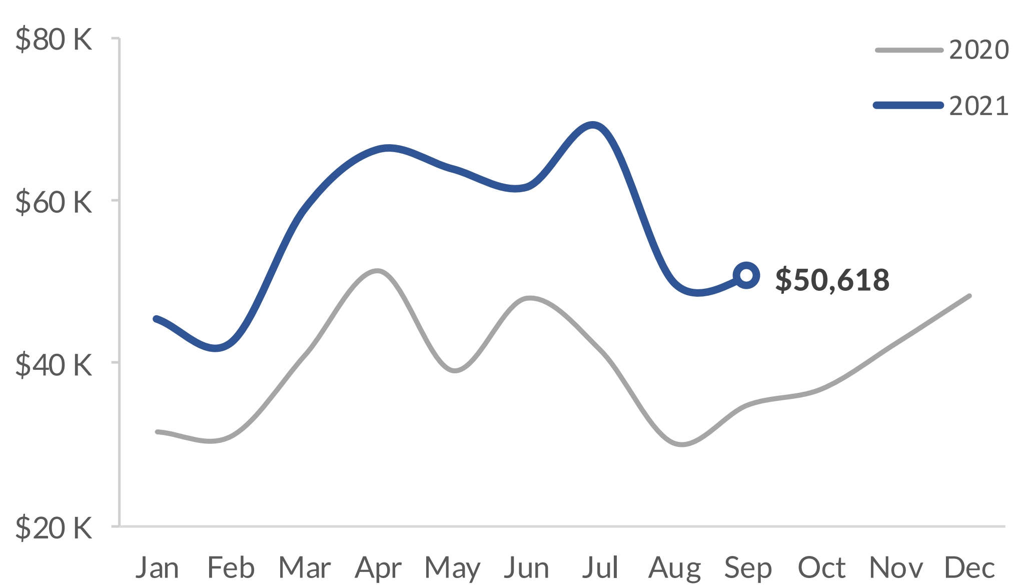
$50,618

Total Monthly Revenue

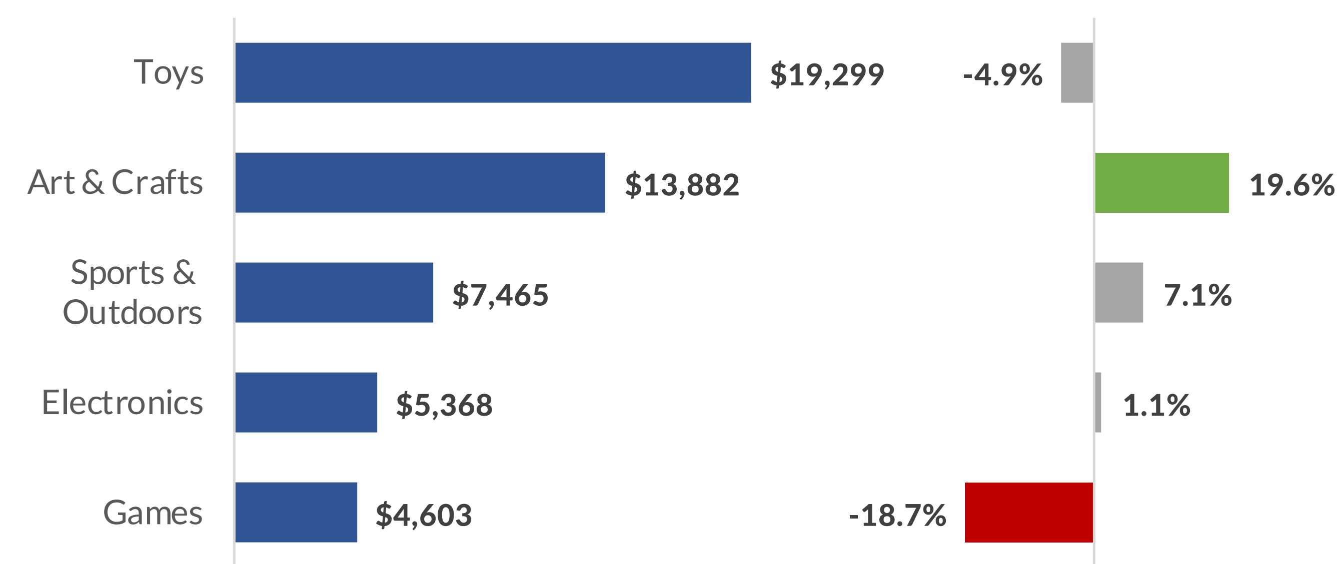
1.6% M-o-M

45.3% Y-o-Y

Total Revenue by Date



Revenue & MoM Revenue % Change by Product Category



Products with 0 Stock

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

$1,640

Potential Monthly Revenue **Loss**

Top & Bottom 5 Products by MoM Revenue Change

Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	$2,011
Toys	Dinosaur Figures	$2,893	$989
Games	Monopoly	$900	$740
Art & Crafts	Magic Sand	$4,589	$688
Art & Crafts	Barrel O' Slime	$1,357	$551

Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

*Copyright Maven Analytics, LLC

STEP 6: TELL A STORY

REGIONAL SALES DASHBOARD

How did the region perform in **September 2021**?

Region: **New York**

MAVEN **TOYS**

This is how much **revenue** we generated...

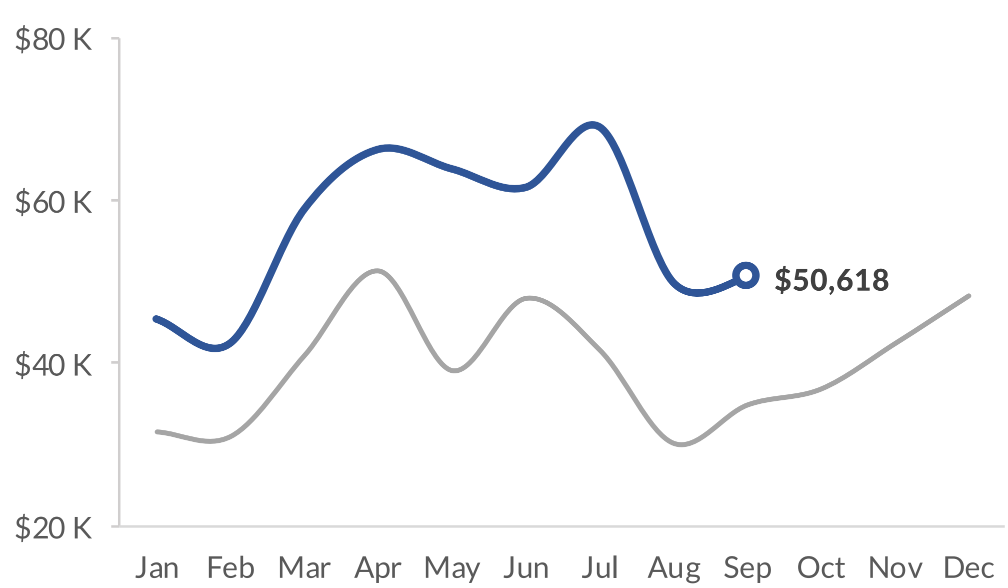
$50,618

Total Monthly Revenue

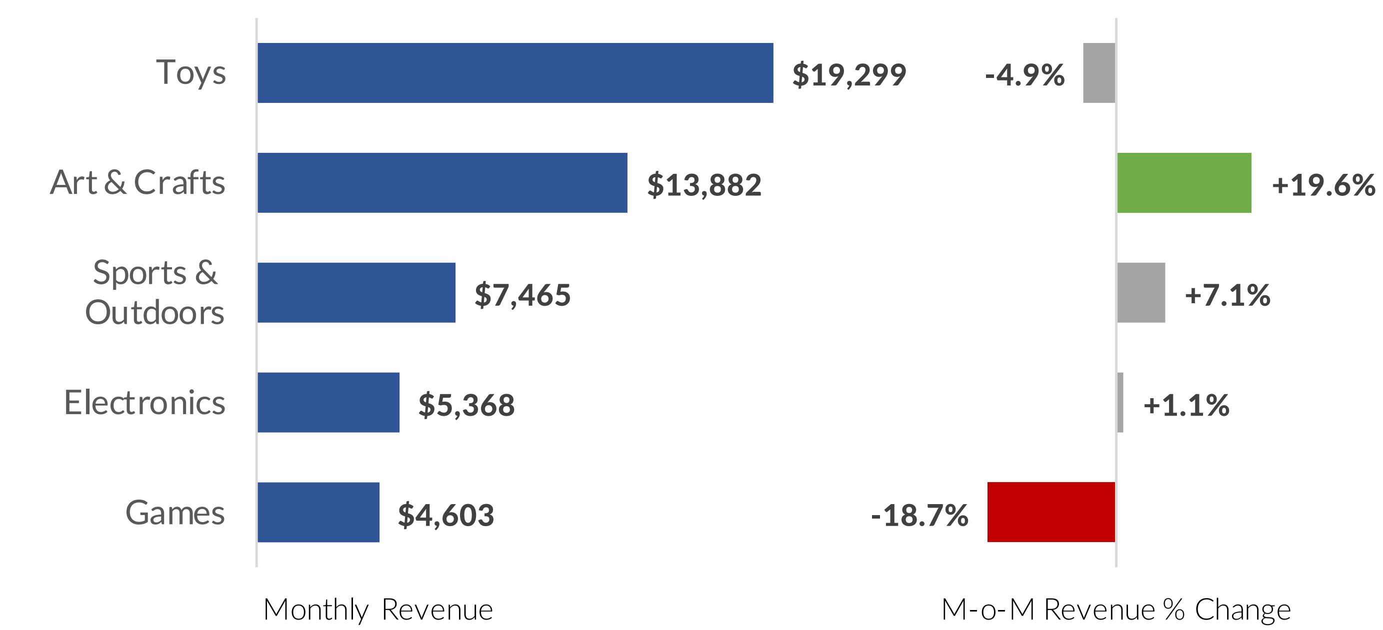
+1.6% M-o-M

+45.3% Y-o-Y

... and the trend in **2021** vs **2020**



This is the revenue split by **product category** this month...



We are **losing sales** here...

Store Name	Product Name	Stock
JFK Airport	Gamer Headphones	0
JFK Airport	Hot Wheels 5-Pack	0
Times Square	Dino Egg	0
Times Square	Playfoam	0

$1,640

Potential Monthly Revenue **Loss**

... where these **5 products** drove the **increases** & **decreases** in revenue

Category	Product	Revenue	Δ Revenue
Art & Crafts	Playfoam	$3,352	+$2,011
Toys	Dinosaur Figures	$2,893	+$989
Games	Monopoly	$900	+$740
Art & Crafts	Magic Sand	$4,589	+$688
Art & Crafts	Barrel O' Slime	$1,357	+$551

Category	Product	Revenue	Δ Revenue
Games	Rubik's Cube	$640	-$1,359
Toys	Lego Bricks	$9,318	-$800
Toys	Mr. Potatohead	$460	-$719
Art & Crafts	Etch A Sketch	$882	-$525
Games	Glass Marbles	$989	-$517

*Copyright Maven Analytics, LLC

KEY TAKEAWAYS

Always answer the **3 key questions** when visualizing data

- *What type of data are you working with? What do you want to communicate? Who is the end user?*

Keep it simple by eliminating noise and distractions

- *"Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away"*

Tell a story by focusing the viewer's attention

- *Use pre-attentive attributes, gestalt principles, and thoughtful layouts to create a clear narrative*

Remember that **context** is key

- *Context is what makes the data mean something, and helps users interpret it accurately*

*Copyright Maven Analytics, LLC